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Homework 4

The goal of this homework is to study the structural analysis by FINITE ELEMENT METHOD of the one dimensional beam with fixed support and square cross section showed in figure 1. The beam is subjected to a sinusoidal load and above there are all the configurations through which it is requested to find the approximated solution that best approximates the analytical one.

Different configurations

A) 1 linear element with 2 nodes
B) 2 linear elements with 3 nodes
C) 3 linear elements with 4 nodes
D) 1 quadratic element with 3 nodes

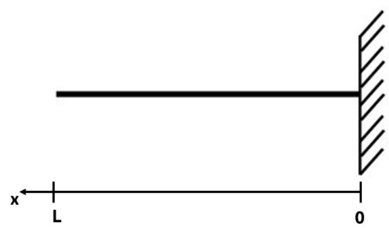

Figure 1: analysed beam

Boundary conditions

$$\begin{cases} x = 0 \longrightarrow N = 0 \\ x = L \longrightarrow u = 0 \end{cases}$$

Applied load

$p(x) = p_0 \; sin(\frac{\pi x}{2L})$

showed in figure 2.

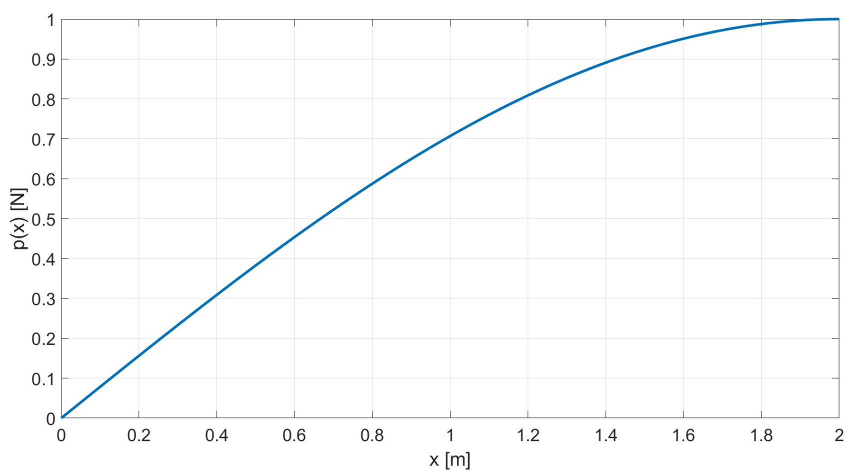

Figure 2: Applied load p(x).

Physical properties of aluminum beam

- $p_0 = 1 \; N/m$
- $L = 2 \; m$
- $t = 0.01 \; m$ and since the beam has square section:
  $L = 1 \; cm$ and $A = 1 \; cm^2$
- $E = 70 \; GPa \; (Al)$
- $\rho = 2780 \; Kg/m^3$
- $\nu = 0.33$

Exact solution

Starting from the constitutive and equilibrium equations it is possible to derive the analytical solution in terms of displacement field, strain field and axial stress field.

$$\begin{cases} N(x) = EA\epsilon = EA\frac{du}{dx} & \Longrightarrow \text{constitutive equation} \\ \frac{dN}{dx} + p(x) = 0 & \Longrightarrow \text{equilibrium equation} \end{cases} \longrightarrow -EA\frac{d^2u}{dx^2} = p_0 sin(\frac{\pi x}{2L}) \tag{1}$$

The eq.1 is a 2^{nd} order ODE and they can be found:

- Homogeneous solution: $-EA\frac{d^2u}{dx^2} = 0 \Longrightarrow u_o(x) = c_1x + c_2$
- Particular solution (using similarity method): $u_p(x) = \bar{u}\ sin(\frac{\pi x}{2L})$
  integrating respect to x the eq.1 twice it is obtained that: $u(x) = \frac{p_0 4L^2}{EA\pi^2} sin(\frac{\pi x}{2L})$
  so $\bar{u} = \frac{p_0 4L^2}{EA\pi^2}$ and $u_p(x) = \frac{p_0 4L^2}{EA\pi^2} sin(\frac{\pi x}{2L})$

so exact solution can be written as: $u(x) = u_p(x) + u_o(x) = \frac{p_0 4L^4}{EA\pi^2} sin(\frac{\pi x}{2L}) + c_1x + c_2$ and imposing boundary conditions they can be found $c_1 = -\frac{p_0 2L}{EA\pi}$ and $c_2 = \frac{p_0 2L^2}{EA\pi}(1 - \frac{2}{\pi})$.

The final exact solutions are:

$$\begin{cases} u(x) = \frac{p_0 2L}{EA\pi}(\frac{2L}{\pi}(sin(\frac{\pi x}{2L} - 1)) - x - L) & \text{displacement field} \\ \epsilon(x) = \frac{du}{dx} = \frac{p_0 2L}{EA\pi}(cos(\frac{\pi x}{2L}) - 1) & \text{strain field} \\ N(x) = EA\epsilon(x) = \frac{p_0 2L}{\pi}(cos(\frac{\pi x}{2L}) - 1) & \text{axial stress field} \end{cases} \tag{2}$$

Finite element method solution

The beam is divided into different number of segments and nodes for each configuration. The approximated displacement is found solving the equation:

$$u(x) = \underline{\underline{H}}\ \underline{\mathrm{U}} \tag{3}$$

where $\underline{\underline{H}} = \{h_i\}$ is the matrix of the shape functions and $\underline{\mathrm{U}}$ is the vector of the nodal displacements. It is calculated imposing assigned load displacements and boundary conditions by solving the equation:

$$\underline{\underline{K}}\ \underline{\mathrm{U}} = \underline{\mathrm{F}} \tag{4}$$

where

$$\underline{\underline{K}} = \int_0^L \underline{\underline{B}}^T \ \underline{\underline{C}} \ \underline{\underline{B}} \ dx \quad \text{is the stiffness matrix} \qquad \text{with} \quad \underline{\underline{B}} = \underline{\underline{D}} \ \underline{\underline{H}} \qquad \underline{\underline{C}} = EA \ \ \text{and}$$

$$\underline{\underline{D}} \ \text{differential operator} \quad \text{and} \ \ \underline{\text{F}} = \int_0^L \underline{\underline{H}}^T \ p(x) \ dx \quad \text{is the vector of nodal forces.}$$

CASE A: Linear element with 2 nodes

Figure 3: Configuration A

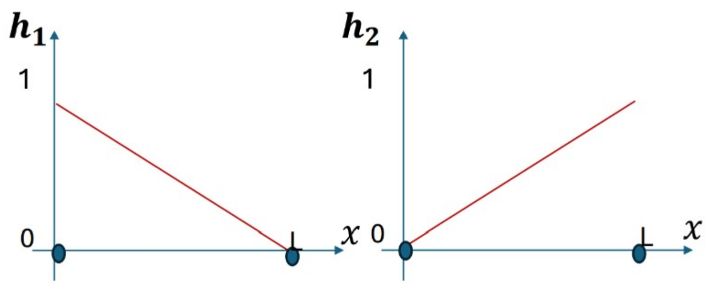

Figure 4: Shape functions for configuration A

The shape functions showed in figure 4 are:

$$h_1(x) = 1 - \frac{x}{L} \qquad\qquad h_2(x) = \frac{x}{L} \tag{5}$$

so the displacement field can be represented as: $\quad u(x) = h_1(x)u_1 + h_2(x)u_2$
and in order to solve the system 4 they are evaluated the stiffness matrix $\underline{\underline{K}}$ and the vector of nodal forces $\underline{\text{F}}$:

$$\underline{\underline{K}} = \int_0^L \begin{bmatrix} -\frac{1}{L} \\ \frac{1}{L} \end{bmatrix} E \ A \begin{bmatrix} -\frac{1}{L} & \frac{1}{L} \end{bmatrix} dx = \frac{EA}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$\underline{\text{F}} = \int_0^L \begin{bmatrix} 1 - \frac{x}{L} \\ \frac{x}{L} \end{bmatrix} \ p_0 \ sin(\frac{\pi x}{2L}) \ dx = \frac{2Lp_0}{\pi} \begin{bmatrix} 1 - \frac{2}{\pi} \\ \frac{2}{\pi} \end{bmatrix}$$

where the last integral is obtained using integration by parts. Then the overall system can be rewritten as:

$$\frac{EA}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} U_1 \\ U_2 \end{bmatrix} = \begin{bmatrix} F_1 \\ R \end{bmatrix}$$

where U_1 and R are the unknowns and U_2 is null because of the boundary conditions of the problem. So F_1 can be evaluated as:

$$F_1 = \int_0^L (1 - \frac{x}{L})\ p_0\ sin(\frac{\pi x}{2L})\ dx = \frac{2Lp_0}{\pi}(1 - \frac{2}{\pi})$$

and then the nodal vector's displacement in the first node is:

$$U_1 = \frac{2L^2}{\pi EA}\ p_0\ (1 - \frac{2}{\pi})$$

Now there are all the useful quantities to compute the displacement field:

$$u_A = \begin{bmatrix} 1 - \frac{x}{L} & \frac{x}{L} \end{bmatrix} \begin{bmatrix} U_1 \\ 0 \end{bmatrix} \longrightarrow u_A(x) = \frac{2L}{\pi EA}\ p_0(L - x)(1 - \frac{2}{\pi})$$

CASE B: 2 linear elements with 3 nodes

Figure 5: Configuration B

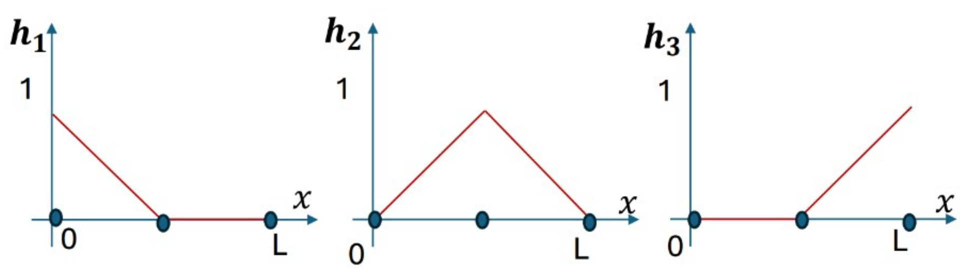

Figure 6: Shape functions for configuration B

The shape functions showed in figure 6 are:

$$h_1(x) = \begin{cases} 1 - \frac{2x}{L} & \text{for} \quad 0 \leq x < \frac{L}{2} \\ 0 & \text{for} \quad \frac{L}{2} \leq x < L \end{cases} \tag{6}$$

$$h_2(x) = \begin{cases} \frac{2x}{L} & \text{for} \quad 0 \leq x < \frac{L}{2} \\ 2(1 - \frac{x}{L}) & \text{for} \quad \frac{L}{2} \leq x < L \end{cases} \tag{7}$$

$$h_3(x) = \begin{cases} 0 & \text{for} \quad 0 \leq x < \frac{L}{2} \\ \frac{2x}{L} - 1 & \text{for} \quad \frac{L}{2} \leq x < L \end{cases} \tag{8}$$

so the displacement field can be represented as: $u(x) = h_1(x)u_1 + h_2(x)u_2 + h_3(x)u_3$ and in order to solve the system 4 they are evaluated the stiffness matrix $\underline{\underline{K}}$ and the vector of nodal forces $\underline{F}$:

$$\underline{\underline{K}} = \int_0^{\frac{L}{2}} \begin{bmatrix} -\frac{2}{L} \\ \frac{2}{L} \\ 0 \end{bmatrix} E\ A \left[-\frac{2}{L}\ \frac{2}{L}\ 0\right] dx \ + \ \int_{\frac{L}{2}}^{L} \begin{bmatrix} 0 \\ -\frac{2}{L} \\ \frac{2}{L} \end{bmatrix} E\ A \left[0\ -\frac{2}{L}\ \frac{2}{L}\right] dx = \frac{2EA}{L} \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$$

$$\underline{F} = \int_0^{\frac{L}{2}} \begin{bmatrix} 1-\frac{2x}{L} \\ \frac{2x}{L} \\ 0 \end{bmatrix} p_0\ sin(\frac{\pi x}{2L})\ dx \ + \ \int_{\frac{L}{2}}^{L} \begin{bmatrix} 0 \\ 2(1-\frac{x}{L}) \\ \frac{2x}{L}-1 \end{bmatrix} p_0\ sin(\frac{\pi x}{2L})\ dx = \frac{Lp_0}{\pi} \begin{bmatrix} 2(1-\frac{2\sqrt{2}}{\pi}) \\ \frac{8}{\pi}(\sqrt{2}-1) \\ \frac{4}{\pi}(2-\sqrt{2}) \end{bmatrix}$$

Then the overall system can be rewritten as:

$$\frac{2EA}{L} \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} U_1 \\ U_2 \\ 0 \end{bmatrix} = \begin{bmatrix} F_1 \\ F_2 \\ R \end{bmatrix}$$

where U_1, U_2 and R are the unknowns.
The solution in terms of U_1 and U_2 is:

$$\begin{cases} U_1 = \frac{L}{2EA}(2F_1 + F_2) \\ U_2 = \frac{L}{2EA}(F_1 + F_2) \end{cases}$$

Now there are all the useful quantities to compute the displacement field:

$$\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 1-\frac{2x}{L} & \frac{2x}{L} & 0 \\ 0 & 2(1-\frac{x}{L}) & \frac{2x}{L}-1 \end{bmatrix} \begin{bmatrix} U_1 \\ U_2 \\ 0 \end{bmatrix}$$

$$\begin{cases} u_B(x) = \frac{p_0 L^2}{2EA\pi}[(1-\frac{2x}{L})(4-\frac{8}{\pi}) + \frac{2x}{L}(2+\frac{4\sqrt{2}}{\pi}-\frac{8}{\pi})] & \text{for} \ \ 0 \le x < \frac{L}{2} \\ u_B(x) = \frac{p_0 L^2}{EA\pi}(1-\frac{x}{L})(2-\frac{8}{\pi}+\frac{4\sqrt{2}}{\pi}) & \text{for} \ \ \frac{L}{2} \le x < L \end{cases}$$

CASE C: 3 linear elements with 4 nodes

Figure 7: Configuration C

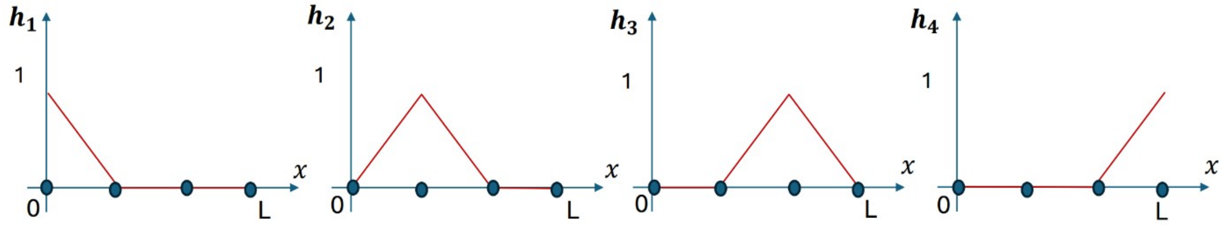

Figure 8: Shape functions for configuration C

The shape functions showed in figure 8 are:

$$h_1(x) = \begin{cases} 1 - \frac{3x}{L} & \text{for } \ 0 \leq x < \frac{L}{3} \\ 0 & \text{for } \ \frac{L}{3} \leq x < \frac{2}{3}L \\ 0 & \text{for } \ \frac{2}{3}L \leq x < L \end{cases} \qquad h_2(x) = \begin{cases} \frac{3x}{L} & \text{for } \ 0 \leq x < \frac{L}{3} \\ 2 - \frac{3x}{L} & \text{for } \ \frac{L}{3} \leq x < \frac{2}{3}L \\ 0 & \text{for } \ \frac{2}{3}L \leq x < L \end{cases} \tag{9}$$

$$h_3(x) = \begin{cases} 0 & \text{for } \ 0 \leq x < \frac{L}{3} \\ \frac{3x}{L} - 1 & \text{for } \ \frac{L}{3} \leq x < \frac{2}{3}L \\ 3 - \frac{3x}{L} & \text{for } \ \frac{2}{3}L \leq x < L \end{cases} \qquad h_4(x) = \begin{cases} 0 & \text{for } \ 0 \leq x < \frac{L}{3} \\ 0 & \text{for } \ \frac{L}{3} \leq x < \frac{2}{3}L \\ \frac{3x}{L} - 2 & \text{for } \ \frac{2}{3}L \leq x < L \end{cases} \tag{10}$$

so the displacement field can be represented as:
$u(x) = h_1(x)u_1 + h_2(x)u_2 + h_3(x)u_3 + h_4(x)u_4$
and in order to solve the system 4 they are evaluated the stiffness matrix $\underline{\underline{K}}$ and the vector of nodal forces $\underline{\text{F}}$:

$$\underline{\underline{K}} = \int_0^{\frac{L}{3}} \begin{bmatrix} -\frac{3}{L} \\ \frac{3}{L} \\ 0 \\ 0 \end{bmatrix} E\ A \begin{bmatrix} -\frac{3}{L} & \frac{3}{L} & 0 & 0 \end{bmatrix} dx \ + \int_{\frac{L}{3}}^{\frac{2}{3}L} \begin{bmatrix} 0 \\ -\frac{3}{L} \\ \frac{3}{L} \\ 0 \end{bmatrix} E\ A \begin{bmatrix} 0 & -\frac{3}{L} & \frac{3}{L} & 0 \end{bmatrix} dx \ +$$
$$+ \int_{\frac{2}{3}L}^{L} \begin{bmatrix} 0 \\ 0 \\ -\frac{3}{L} \\ \frac{3}{L} \end{bmatrix} E\ A \begin{bmatrix} 0 & 0 & -\frac{3}{L} & \frac{3}{L} \end{bmatrix} dx = \frac{3EA}{L} \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{bmatrix}$$

$$\underline{\text{F}} = \int_0^{\frac{L}{3}} \begin{bmatrix} 1 - \frac{3x}{L} \\ \frac{3x}{L} \\ 0 \\ 0 \end{bmatrix} \ p_0\ sin(\frac{\pi x}{2L})\ dx \ + \int_{\frac{L}{3}}^{\frac{2}{3}L} \begin{bmatrix} 0 \\ 2 - \frac{3}{L}x \\ \frac{3x}{L} - 1 \\ 0 \end{bmatrix} \ p_0\ sin(\frac{\pi x}{2L})\ dx \ +$$
$$+ \int_{\frac{2}{3}L}^{L} \begin{bmatrix} 0 \\ 0 \\ 3 - \frac{3x}{L} \\ \frac{3x}{L} - 2 \end{bmatrix} \ p_0\ sin(\frac{\pi x}{2L})\ dx = \frac{Lp_0}{\pi} \begin{bmatrix} 2 - \frac{6}{\pi} \\ \frac{12}{\pi} - \frac{6\sqrt{3}}{\pi} \\ \frac{12\sqrt{3}}{\pi} - \frac{18}{\pi} \\ \frac{12}{\pi} - \frac{6\sqrt{3}}{\pi} \end{bmatrix}$$

Then the overall system can be rewritten as:

$$\frac{3EA}{L}\begin{bmatrix}1 & -1 & 0 & 0\\ -1 & 2 & -1 & 0\\ 0 & -1 & 2 & -1\\ 0 & 0 & -1 & 1\end{bmatrix}\begin{bmatrix}U_1\\ U_2\\ U_3\\ 0\end{bmatrix}=\begin{bmatrix}F_1\\ F_2\\ F_3\\ R\end{bmatrix}$$

where U_1, U_2, U_3 and R are the unknowns.
The solution in terms of U_1, U_2 and U_3 is:

$$\begin{cases}U_1 = \frac{L}{3EA}(3F_1+2F_2+F_3)\\ U_2 = \frac{L}{3EA}(2F_1+2F_2+F_3)\\ U_3 = \frac{L}{3EA}(F_1+F_2+F_3)\end{cases}$$

Now there are all the useful quantities to compute the displacement field:

$$\begin{bmatrix}u_1\\ u_2\\ u_3\end{bmatrix}=\begin{bmatrix}1-\frac{3x}{L} & \frac{3x}{L} & 0 & 0\\ 0 & 2-\frac{3x}{L} & \frac{3x}{L}-1 & 0\\ 0 & 0 & 3-\frac{3x}{L} & \frac{3x}{L}-2\end{bmatrix}\begin{bmatrix}U_1\\ U_2\\ U_3\\ 0\end{bmatrix}$$

$$\begin{cases}u_C(x) = (1-\frac{3x}{L})\frac{L^2p_0}{3EA\pi}(6-\frac{12}{\pi})+\frac{Lp_0x}{EA\pi}(4-\frac{6}{\pi}) & \text{for } 0\leq x<\frac{L}{3}\\ u_C(x) = (2-\frac{3x}{L})\frac{L^2p_0}{3EA\pi}(4-\frac{6}{\pi})+(\frac{3x}{L}-1)\frac{L^2p_0}{3EA\pi}(2-\frac{12}{\pi}+\frac{6\sqrt{3}}{\pi}) & \text{for } \frac{L}{3}\leq x<\frac{2}{3}L\\ u_C(x) = (3-\frac{3x}{L})\frac{L^2p_0}{3EA\pi}(2-\frac{12}{\pi}+\frac{6\sqrt{3}}{\pi}) & \text{for } \frac{2}{3}L\leq x<L\end{cases}$$

CASE D: 1 quadratic element with 3 nodes

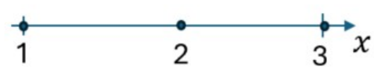

Figure 9: Configuration D

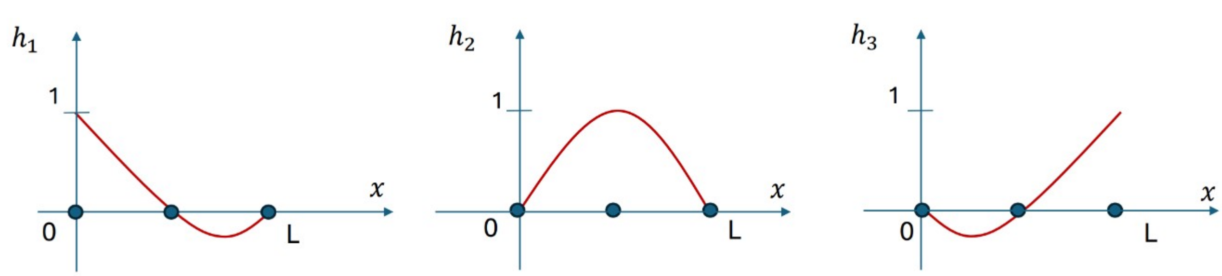

Figure 10: Shape functions for configuration D

The shape functions showed in figure 10 are:

$$\begin{cases} h_1(x) = \frac{2x^2}{L^2} - \frac{3x}{L} + 1 \\ h_2(x) = -\frac{4x^2}{L^2} + \frac{4x}{L} \\ h_3(3) = \frac{2x^2}{L^2} - \frac{x}{L} \end{cases} \tag{11}$$

so the displacement field can be represented as: $u(x) = h_1(x)u_1 + h_2(x)u_2 + h_3(x)u_3$
and in order to solve the system 4 they are evaluated the stiffness matrix $\underline{\underline{K}}$ and the vector of nodal forces $\underline{\mathrm{F}}$:

$$\underline{\underline{K}} = \int_0^L \begin{bmatrix} \frac{4x}{L} - \frac{3}{L} \\ \frac{-8x}{L^2} + \frac{4}{L} \\ \frac{4x}{L^2} - \frac{1}{L} \end{bmatrix} E\ A \begin{bmatrix} \frac{4x}{L} - \frac{3}{L} & \frac{-8x}{L^2} + \frac{4}{L} & \frac{4x}{L^2} - \frac{1}{L} \end{bmatrix} dx = \frac{EA}{3L} \begin{bmatrix} 7 & -8 & 1 \\ -8 & 16 & -8 \\ 1 & -8 & 7 \end{bmatrix}$$

$$\underline{\mathrm{F}} = \int_0^L \begin{bmatrix} \frac{2x^2}{L^2} - \frac{3x}{L} + 1 \\ \frac{-4x^2}{L^2} + \frac{4x}{L} \\ \frac{2x^2}{L^2} - \frac{x}{L} \end{bmatrix} p_0\ sin(\frac{\pi x}{2L})\ dx = \frac{2Lp_0}{\pi} \begin{bmatrix} 1 + \frac{2}{\pi} - \frac{16}{\pi^2} \\ \frac{32}{\pi^2} - \frac{8}{\pi} \\ \frac{6}{\pi} - \frac{16}{\pi^2} \end{bmatrix}$$

Then the overall system can be rewritten as:

$$\frac{EA}{3L} \begin{bmatrix} 7 & -8 & 1 \\ -8 & 16 & -8 \\ 1 & -8 & 7 \end{bmatrix} \begin{bmatrix} U_1 \\ U_2 \\ 0 \end{bmatrix} = \begin{bmatrix} F_1 \\ F_2 \\ R \end{bmatrix}$$

where U_1, U_2 and R are the unknowns.
The solution in terms of U_1 and U_2 is:

$$\begin{cases} U_1 = \frac{L}{EA}(F_1 + \frac{F_2}{2}) \\ U_2 = \frac{L}{2EA}(\frac{7}{8}F_2 + F_1) \end{cases}$$

Now there are all the useful quantities to compute the displacement field:

$$u = \begin{bmatrix} \frac{2x^2}{L^2} - \frac{3x}{L} + 1 & \frac{-4x^2}{L^2} + \frac{4x}{L} & \frac{2x^2}{L^2} - \frac{x}{L} \end{bmatrix} \begin{bmatrix} U_1 \\ U_2 \\ 0 \end{bmatrix}$$

$$u_D(x) = (\frac{2x^2}{L^2} - \frac{3x}{L} + 1)\frac{2L^2p_0}{EA\pi}(1 - \frac{2}{\pi}) + (-\frac{4x^2}{L^2} + \frac{4x}{L})\frac{L^2p_0}{EA\pi}(\frac{12}{\pi^2} - \frac{5}{\pi} + 1)$$

Results

In figure 11 there are the displacement field of the analytical solution and all the four studied FEM cases. From fig.11 it is possible to evaluate which is the best approximated solution of

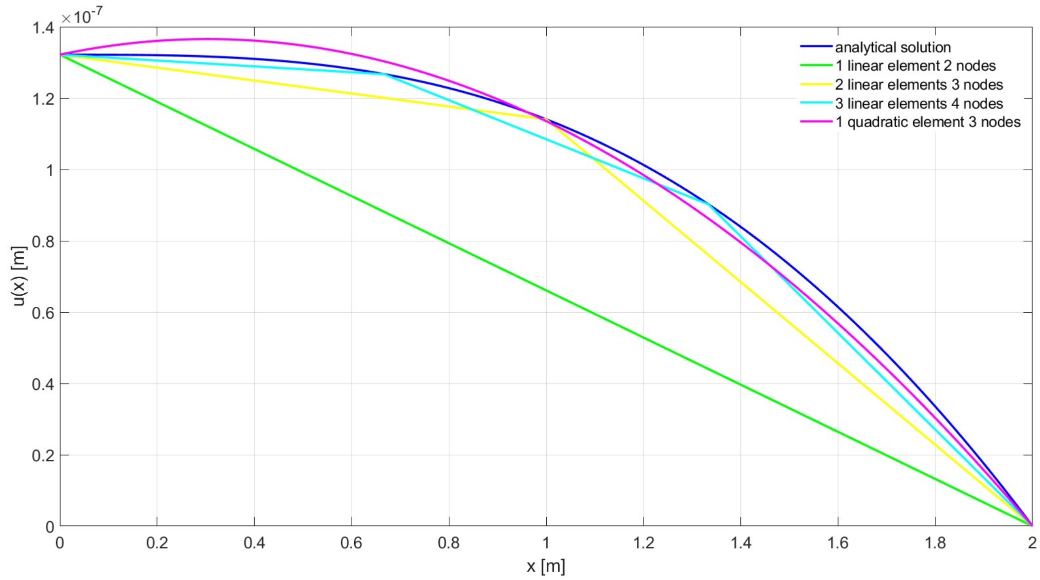

Figure 11: Comparison between the displacement field of the analytical solution and FEM discretization configurations

the displacement field. The one that comes closest to the analytical solution is the configuration D, which has one quadratic element and 3 nodes. Moreover, it is possible to notice that the convergence is reached faster when the number of elements of the configuration is increased or when the order of the shape functions is increasing.

Table 1 shows the displacement in the middle of the beam, at the beginning of it and the relative percentage error computed as the difference between the subtended area of the analytical solution and the specific case. As expected $u(0)$ is equal for each case because it is related with the boundary condition on the axial force N. The relative error decreases until the best value of approximately zero for the quadratic case.

One can evaluate the convergence of the strain field among all the cases using the compatibility equation $\epsilon(x) = \frac{du}{dx}$:

	$\mathbf{u(L/2)}$	$\mathbf{u(0)}$	**Relative error**
Analytical	$\frac{p_0 2L^2}{EA\pi}(\frac{\sqrt{2}-2}{\pi}+\frac{1}{2})$	$\frac{p_0 2L^2}{EA\pi}(1-\frac{2}{\pi})$	0%
Case A	$\frac{p_0 L^2}{\pi EA}(1-\frac{2}{\pi})$	$\frac{p_0 2L^2}{\pi EA}(1-\frac{2}{\pi})$	32.45%
Case B	$\frac{p_0 L^2}{2EA\pi}(2-\frac{8}{\pi}+\frac{4\sqrt{2}}{\pi})$	$\frac{p_0 2L^2}{\pi EA}(1-\frac{2}{\pi})$	7.91%
Case C	$\frac{p_0 L^2}{6EA\pi}(6-\frac{18}{\pi}+\frac{6\sqrt{3}}{\pi})$	$\frac{p_0 2L^2}{\pi EA}(1-\frac{2}{\pi})$	3.54%
Case D	$\frac{p_0 L^2}{EA\pi}(\frac{12}{\pi^2}-\frac{5}{\pi}+1)$	$\frac{p_0 2L^2}{\pi EA}(1-\frac{2}{\pi})$	0%

Table 1: Comparison between all the studied configuration's displacements in terms of: displacement in the middle, displacement at the beginning and relative error with respect to the analytical solution.

Analytical solution: $\epsilon(x) = \dfrac{p_0 2L}{EA\pi}(cos(\dfrac{\pi x}{2L}-1))$

Case A: $\epsilon_A(x) = \dfrac{2Lp_0}{\pi EA}(\dfrac{2}{\pi}-1)$

Case B: $\begin{cases} \epsilon_B(x) = \frac{2p_0 L}{EA\pi}(\frac{2\sqrt{2}}{\pi}-1) & \text{for } \ 0 \le x < \frac{L}{2} \\ \epsilon_B(x) = \frac{2p_0 L}{EA\pi}(\frac{4}{\pi}-\frac{2\sqrt{2}}{\pi}-1) & \text{for } \ \frac{L}{2} \le x < L \end{cases}$

Case C: $\begin{cases} \epsilon_C(x) = \frac{Lp_0}{EA\pi}(\frac{6}{\pi}-2) & \text{for } \ 0 \le x < \frac{L}{3} \\ \epsilon_C(x) = \frac{Lp_0}{EA\pi}(\frac{6\sqrt{3}}{\pi}-\frac{6}{\pi}-2) & \text{for } \ \frac{L}{3} \le x < \frac{2}{3}L \\ \epsilon_C(x) = \frac{Lp_0}{EA\pi}(\frac{12}{\pi}-\frac{6\sqrt{3}}{\pi}-2) & \text{for } \ \frac{2}{3}L \le x < L \end{cases}$

Case D: $\epsilon_D(x) = \dfrac{2Lp_0}{EA\pi}(\dfrac{12x}{L^2\pi}-\dfrac{1}{L}-\dfrac{4}{L\pi}+\dfrac{24}{L\pi^2}-\dfrac{48x}{L^2\pi^2})$

As for the displacement field, also the strain field is approximated in the best way by the last configuration. This is shown in figure 12 where all the configurations have a certain error with respect to the analytical solution and this is obvious because the expressions of $\epsilon(x)$ are constants or linear at most, while the analytical solution has a cosinusoidal trend.

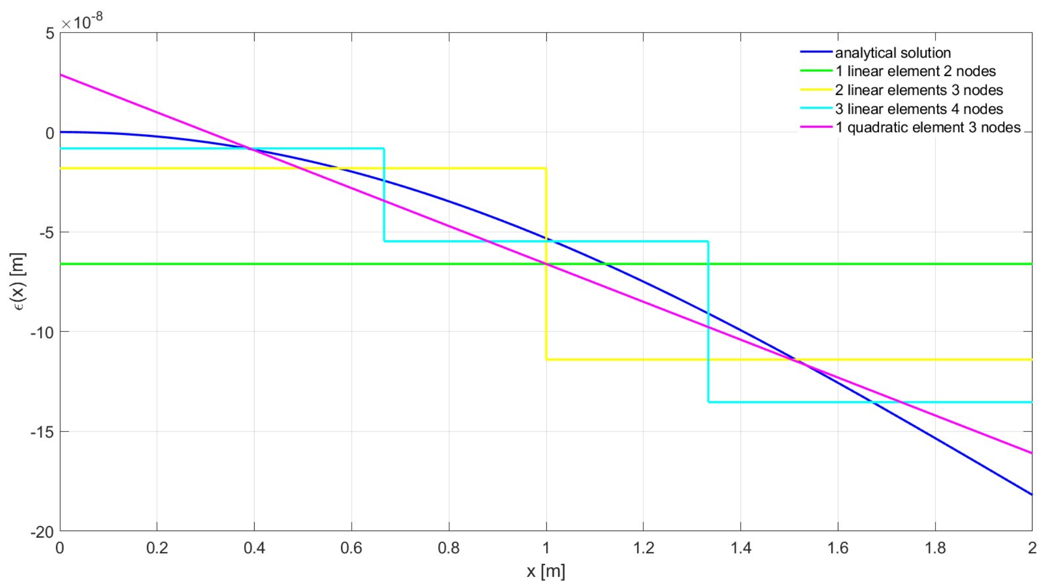

Figure 12: Comparison between the strain field of the analytical solution and FEM discretization configurations.